

BraTS MRI Reconstruction

BME AI Project 1

第 4 组 / 唐志昊 2022533131

2026-04-28

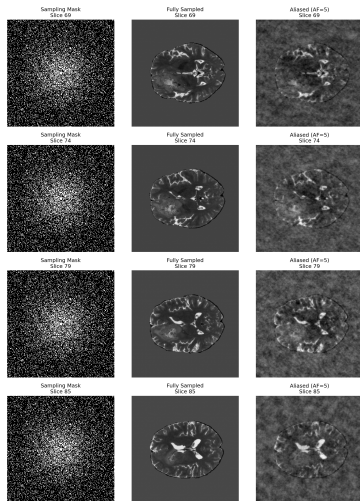
Project Motivation

- Faster MRI acquisition is clinically desirable but undersampling causes aliasing artifacts.
- Goal: reconstruct high-quality T2 MRI from $AF=5$ undersampled k-space.
- Tasks: simulation, baseline reconstruction, and multi-modal unrolled reconstruction.

Dataset and Experimental Setup

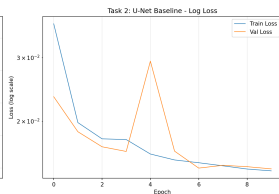
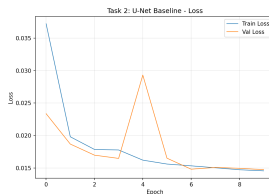
- Dataset: BraTS, using co-registered T1 and T2 modalities.
- Split: patient-level split to avoid train/test leakage.
- Final formal run: 16 central slices per patient, all available patients.
- Runtime optimization: slice caching, pinned memory, TF32, non-blocking transfer.

Task 1: Undersampling Simulation

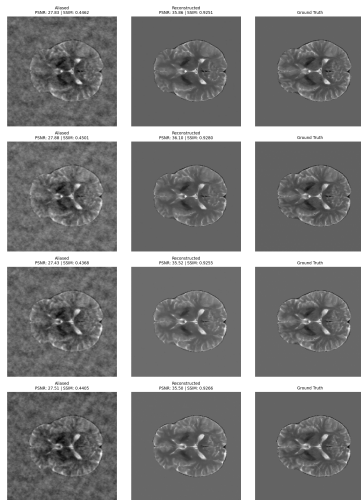


Task 2: Baseline U-Net

- Input: undersampled T2 slice.
- Model: U-Net baseline.
- Loss: MSE.
- Final performance: PSNR 35.18, SSIM 0.9271.

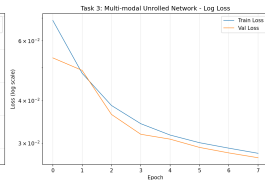
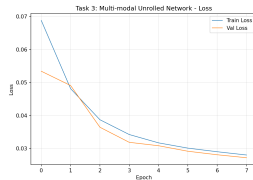


Task 2 Reconstruction Examples



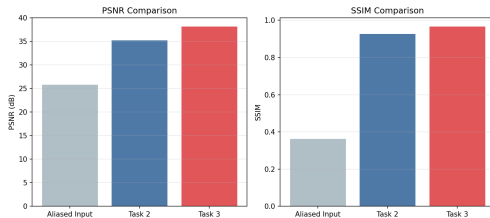
Task 3: Multi-modal Unrolled Network

- Input: undersampled T2 + fully sampled T1.
- Architecture: 2-cascade unrolled U-Net with data consistency.
- Loss: hybrid L1/L2.
- Final performance: PSNR 38.12, SSIM 0.9660.

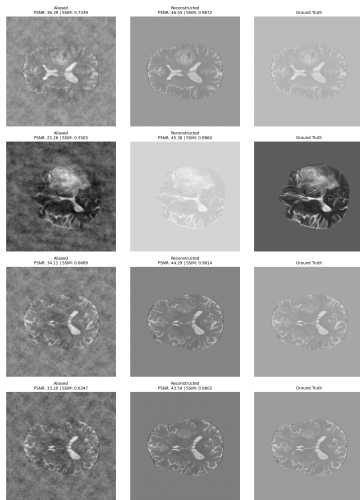


Quantitative Comparison

- Task 3 improves over Task 2 by 2.93 dB PSNR.
- Task 3 improves over Task 2 by 0.0389 SSIM.
- Multi-modal guidance and data consistency both contribute measurable gains.

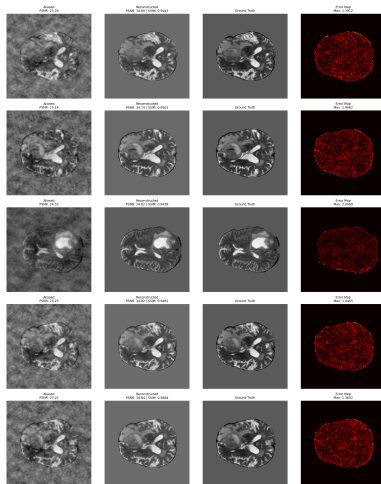


Task 3 Best Reconstructions



Task 3 Error Analysis

5 Worst Reconstruction Cases



Conclusions

- The baseline model already removes most aliasing artifacts effectively.
- The multi-modal unrolled model provides the best fidelity and visual quality.
- Remaining hard cases are mainly complex slices with challenging local structure.
- Future work: stronger edge-aware loss, deeper cascades, and targeted hard-case training.

AI Usage Declaration

- AI tools were used only for debugging support, automation, and draft editing.
- All model choices, experiment validation, metric checks, and final submitted materials were manually reviewed and confirmed.
- This declaration is included to satisfy the assignment requirement.

Thank You

Questions are welcome.